

Division Three Ways

Answer Key

Grade 3

Quick Answers

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|------------------------------|---------------------------|
| 1. pretzels each = 5, — | 7. books per shelf = 9, — |
| 2. vases = 3, — | 8. quotient = 5, — |
| 3. pairs = 8, — | 9. groups = 4, — |
| 4. dots in each group = 5, — | 10. bunches = 5, — |
| 5. quotient = 3, — | 11. quotient = 9, — |
| 6. dots in each group = 6, — | 12. (b) = 6, — |
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What is verified

0 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.OA.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

3.OA.A.3 – problems 1, 2, 3, 7, 10, 11, 12

Difficulty 1–4 of 5 across 24 tagged checks.

Common wrong answers

12. *If they got 20: subtracted the bowls from the berries instead of dividing*

1. Two friends share 10 pretzels equally.

Solution: Sharing 10 into 2 equal groups is the equation $10 \div 2 = 5$. Each friend gets 5 pretzels, because $2 \times 5 = 10$. 5

2. 18 flowers, 6 flowers in each vase.

Solution: Making groups of 6 from 18 is $18 \div 6 = 3$. The florist fills 3 vases, because $6 \times 3 = 18$. 3

3. 16 socks sorted into pairs of 2.

Solution: Making groups of 2 from 16 is $16 \div 2 = 8$. Sam makes 8 pairs, because $2 \times 8 = 16$. 8

4. Drawing: 15 dots in 3 equal groups.

Solution: The drawing splits 15 into 3 equal groups, which is $15 \div 3 = 5$. Each group holds 5 dots — count one group to check. 5

5. A division story for $12 \div 4$, then solve.

Solution: Any story with 12 things and 4 groups works, e.g. “12 muffins are shared equally onto 4 plates. How many muffins on each plate?” (Groups of 4 also work: “12 muffins packed 4 to a box — how many boxes?”) The quotient: $12 \div 4 = 3$, because $4 \times 3 = 12$. 3

6. Drawing: 24 dots in 4 equal groups.

Solution: Splitting 24 into 4 equal groups is $24 \div 4 = 6$. Each group holds 6 dots, because $4 \times 6 = 24$. 6

7. 27 books shared equally on 3 shelves.

Solution: Sharing 27 into 3 equal groups is $27 \div 3 = 9$. Each shelf holds 9 books, because $3 \times 9 = 27$. 9

8. A division story for $30 \div 6$, then solve.

Solution: Example story: “30 crayons are shared equally among 6 friends. How many crayons each?” The quotient: $30 \div 6 = 5$, because $6 \times 5 = 30$. 5

9. Drawing: 20 dots circled in groups of 5.

Solution: Circling groups of 5 out of 20 is $20 \div 5 = 4$. There are 4 groups, because $5 \times 4 = 20$. 4

10. 40 balloons tied into bunches of 8.

Solution: Making groups of 8 from 40 is $40 \div 8 = 5$. The clown makes 5 bunches, because $8 \times 5 = 40$. 5

11. A division story for $36 \div 4$, then solve.

Solution: Example story: “36 stickers are shared equally onto 4 pages. How many stickers on each page?” The quotient: $36 \div 4 = 9$, because $4 \times 9 = 36$. 9

12. Challenge: Maya shares 24 berries among 4 bowls; the drawing shows 4 equal groups.

Solution (a): The matching equation is $24 \div 4$ — 24 berries split into 4 equal bowls, exactly what the drawing shows. $4 \div 24$ is backwards (it would split 4 things among 24 bowls), and $24 - 4$ takes 4 berries away instead of sharing them all.

Solution (b): $24 \div 4 = 6$ berries in each bowl, because $4 \times 6 = 24$. Subtracting instead ($24 - 4 = 20$) does not share anything. 6